# 1 Inverse Trigonometry and Identities

**Central question:** How can we recover an angle from its trigonometric value, and how do we prove two expressions represent the same function?

## 1.1 An inverse is not a reciprocal

The notation $\sin^{-1} x$ means the inverse function $\arcsin x$. It does not mean $1/\sin x = \csc x$. The inverse answers an angle question; the reciprocal answers a ratio question.

Sine is periodic, so an inverse function requires choosing an interval on which sine is one-to-one. We use $[-\pi/2, \pi/2]$. In exactly the same way, cosine uses $[0, \pi]$ and tangent uses $(-\pi/2, \pi/2)$.

| $f(x)$ | Domain | Range |
|---|---|---|
| $\arcsin x$ | $[-1, 1]$ | $[-\pi/2, \pi/2]$ |
| $\arccos x$ | $[-1, 1]$ | $[0, \pi]$ |
| $\arctan x$ | $\mathbb{R}$ | $(-\pi/2, \pi/2)$ |

**Example 1.1.** $\arcsin(1/2) = \pi/6$, $\arccos(-1/2) = 2\pi/3$, and $\arctan(-1) = -\pi/4$.

**Example 1.2.** $\arcsin(\sin(5\pi/6)) = \pi/6$, not $5\pi/6$, because the inverse reports its principal value.

## 1.2 Compositions and triangle geometry

The identity $\sin(\arcsin x) = x$ holds for $-1 \le x \le 1$, but $\arcsin(\sin\theta) = \theta$ only for $\theta$ in the principal range. Suppose $\theta = \arctan(3/4)$. Then a right triangle has legs $3, 4$ and hypotenuse $5$, so

$$\sin(\arctan(3/4)) = \frac{3}{5}.$$

## 1.3 An identity is true throughout its domain

From the unit-circle equation we obtain

$$\boxed{\sin^2 x + \cos^2 x = 1}.$$

Dividing by $\cos^2 x$ (only where $\cos x \neq 0$) gives $1 + \tan^2 x = \sec^2 x$. Dividing by $\sin^2 x$ gives $1 + \cot^2 x = \csc^2 x$ where $\sin x \neq 0$.

**Example 1.3.** For $\sin x \neq 0$,

$$\frac{1 - \cos^2 x}{\sin x} = \frac{\sin^2 x}{\sin x} = \sin x.$$

This is an algebraic proof on the shared domain, not a numerical check at a few inputs.

## 1.4 Angle addition and double angles

Two successive plane rotations produce the sum identities

$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta,$$
$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta.$$

Set $\beta = -\gamma$ for difference identities and $\beta = \alpha$ for double-angle identities:

$$\sin(2x) = 2\sin x\cos x, \qquad \cos(2x) = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x.$$

**Example 1.4.** $\sin(75^\circ) = \sin(45^\circ + 30^\circ) = (\sqrt{6} + \sqrt{2})/4$.

### 1.5 Applications: measured angles

An observation of a tower from 40 m away gives $\tan\theta = 3/4$. The elevation angle is $\arctan(3/4) \approx 36.9°$, while the height is exactly $40(3/4) = 30$ m. Inverse functions recover angles; ratios can often recover distances directly.

### 1.6 Exercises

1. Evaluate exactly $\arcsin(-\sqrt{2}/2)$ and $\arctan(\sqrt{3})$.
2. Explain why $\arcsin(\sin(3\pi/4)) \neq 3\pi/4$.
3. Find $\cos(\arcsin(5/13))$ with a triangle.
4. Find $\sin(\arctan(-2))$ exactly, including its sign.
5. Prove $(\sec^2 x - 1)/\tan x = \tan x$ on its domain.
6. Prove $(\sin x + \cos x)^2 = 1 + \sin(2x)$.
7. Calculate $\cos(75°)$ exactly by angle addition.
8. Express $\cos(2x)$ using $\sin x$ alone.